

Homework # 7. DUE April 13, 2025

1. find the tunneling probability for a STM (Scanning Tunneling Microscope), if the average barrier work function is 4.0 eV, the gap distance is $x_1 = 0.50 \text{ nm}$
2. find the tunneling probability for a STM if the average barrier work function is 4.0 eV, the gap distance is $x_2 = 0.45 \text{ nm}$
3. Compare the STM current in both cases. Hint : The STM current $\sim$ tunneling probability